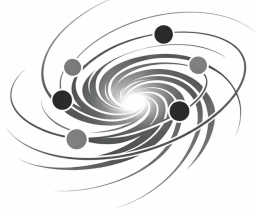


Part of the Monistic Continuum Model (MCM)



Unified Physical Correspondence (UPC)

Structural Correspondence Between MCM Vortices, Quantum Mechanics, and General Relativity

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$$1 = \bigcup_{i=1}^N m_i, \quad m_i \subseteq 1.$$

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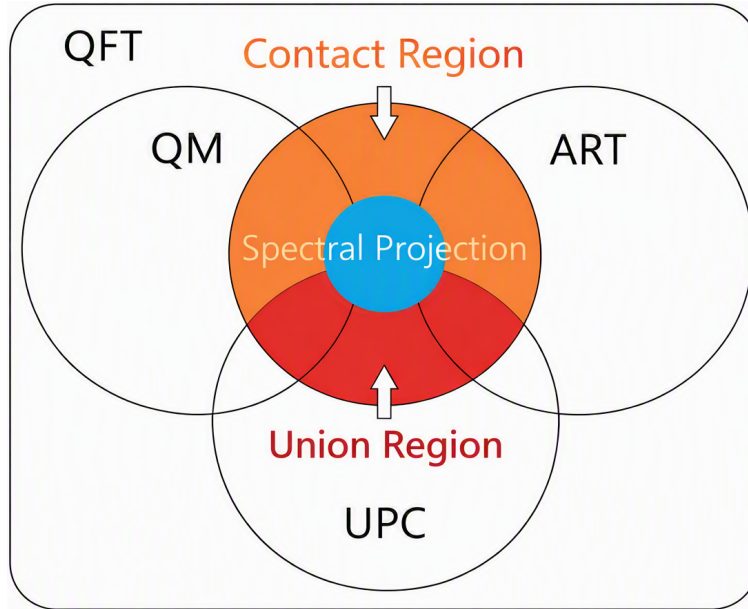
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1 Motivation and Theoretical Framework

The Unified Physical Correspondence (UPC) arises from the necessity to reconcile two fundamental physical descriptions—the quantum mechanical and the relativistic—on a common, operator-based foundation. The UPC provides a natural structure for this purpose, since both quantum fields and metric curvature fields emerge from the same vortex architecture at the Planck scale. The correspondence is therefore not an external postulate but an intrinsic property of the theory.

Unified Physical Correspondence (with QFT Layer)



The Monistic Continuum Model (MCM) provides the ontological framework in which this structure is embedded. The MCM postulates a unified continuum whose internal stress and dynamic distributions form the basis of all physical phenomena. The discrete vortex cells—introduced in earlier formulations of the continuum—carry geometric, topological, and dynamic parameters that operationalize the physical organization of the medium. Within the UPC, these vortex cells constitute the operator-based realization of the MCM.

This operator-based interpretation is consistent with modern operator-geometric frameworks such as Bény [1], who demonstrates that geometric and physical structures can emerge directly from the algebraic properties of operators rather than from a predefined continuum. This provides external support for the UPC/MCM assumption that geometry is not fundamental but operator-derived. The uniformity of the vortex basis constitutes the origin of both limiting cases: the spectral modes of vortex dynamics project onto continuous quantum fields, while the topological density of vortex configurations maps onto curvature fields and metric structures. Both projections originate from

the same discrete vortex architecture and differ only in the type of operators and the chosen projection plane. This ensures a consistent and mathematically well-defined connection between QM and ART.

The idea that both spectral and geometric information can be encoded in a unified operator structure aligns with the operator-algebraic perspective of Bény [1], who shows that spectral properties of operators naturally give rise to geometric features. This reinforces the UPC claim that quantum and relativistic structures emerge from different projections of the same operator basis. The originality of this correspondence lies in the fact that it is formulated entirely within the UPC/MCM structure and requires no external correspondence theories, dualities, or additional quantization procedures. UPC thus provides an intrinsic, operator-based, and geometrically motivated correspondence that derives both the quantum mechanical and relativistic descriptions from a common fundamental architecture.

This intrinsic operator-based correspondence is further supported by emergent-gravity frameworks such as Padmanabhan [2] and Jacobson [3], who demonstrate that gravitational dynamics and Einstein's equations can arise from thermodynamic or informational principles rather than from a fundamental geometric substrate. These insights align naturally with the UPC perspective, in which curvature and metric structure emerge from the topological and dynamical organization of vortex states.

2 Fundamental Basis of MCM and UPC

The fundamental basis of UPC arises from the structural interlocking of two levels: the Monistic Continuum Model (MCM) as the ontological foundation and the discrete vortex architecture as the operator-based realization of this foundation. Both layers share the assumption that physical structures do not emerge from a predefined continuum, but from the internal organization of a unified medium whose stress and dynamic distributions form the basis of all observable phenomena. This perspective aligns with operator-geometric approaches such as Bény [1], who demonstrates that geometric and physical structures can arise from operator algebras rather than from a predefined background manifold.

2.1 Basic Assumptions of the MCM.

The MCM postulates a continuous underlying medium whose internal stress $\sigma(x)$ and dynamic distribution $\mathcal{D}(x)$ determine physical organization. This medium possesses no predefined metric structure; instead, metric properties arise as derived quantities from the local organization of the medium. The fundamental assumptions of the MCM are:

- The physical universe consists of a unified continuum.
- Stress and dynamics are primary quantities from which all structures emerge.
- Geometry, curvature, and fields are derived manifestations of the stress and dynamic distributions.

These assumptions form the ontological foundation on which the UPC operates. The idea that geometry is not fundamental but emergent is consistent with operator-based geometric frameworks such as Bény [1], who shows that geometric information can be encoded in the algebraic structure of operators.

2.2 The MCM-Medium and its Distinction from the Historical Aether

The Monistic Continuum Model (MCM) is ontologically grounded in a universal continuum referred to as the *MCM-medium*. This medium forms the basis of all emergent structures, operators, and correspondences within the MCM architecture. To avoid conceptual misunderstandings, a clear distinction from the historical notion of the aether is required.

The classical aether of the 19th century was understood as a *mechanical medium*. It was assumed to carry electromagnetic waves and to possess material properties such as density, elasticity, pressure, and inertia. This mechanical interpretation proved untenable and was completely abandoned by modern physics.

The MCM-medium differs fundamentally from this historical concept. It is *non-mechanical*: it possesses neither mass nor density, no elasticity, no pressure, and no friction. It is not a material substance and therefore not an aether in the classical sense. Instead, it constitutes an ontological continuum that provides the structural foundation for emergent operators, states, and geodesic organization.

Modern physical theories likewise employ non-mechanical continua, such as quantum fields, the quantum vacuum, or the dynamical structure of spacetime. The MCM-medium belongs to this modern category: it is a universal, non-mechanical carrier continuum that fulfills the conceptual role historically attributed to the aether—yet without any of its mechanical assumptions and fully compatible with the ontological foundations of the MCM.

2.3 Fundamental Objects of the UPC Vortex Architecture.

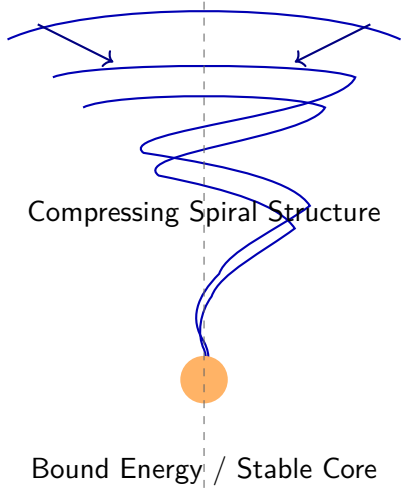
The UPC concretizes the MCM by employing discrete vortex cells at the Planck scale. Each cell c_i carries a state $s_i \in H$ that includes geometric parameters, topological densities, dynamic variables, and spectral amplitudes. The fundamental objects of the UPC vortex architecture are:

- **Vortex cells** as discrete carriers of physical structure,
- **Vortex operators** G, T, D, S, C , which define geometry, topology, dynamics, stability, and cluster formation,
- **Vortex states** $W_n = \{s_i\}_{i \in I_n}$ as physical configurations of the medium.

These objects are mathematically well-defined and consistent with the axioms and operator structures introduced in earlier discrete vortex formulations (e.g., PVT Version 03). The operator-based definition of geometry and dynamics resonates with the spectral-geometric viewpoint of Bombelli [4], where geometric information is encoded in the spectral properties of operators.

UPC–Matter Vortex

Open Modes / High Degrees of Freedom



Geometric Parameters: curvature, spiral pitch, compression
 Topological Charge q , winding number, knot index
 Spectral Modes λ_k , spectral gap Δ
 Local dynamics $d(s_i, \{s_j\})$, interaction structure
 Stability norm $S(W_n)$, core stability
 Cluster operator C : matter aggregation
 MCM embedding: stress $\sigma(x)$, dynamics $D(x)$
 Emergent continuum: QM/GR projections

2.4 Common Carrier Space.

The carrier space of the UPC vortex architecture is a discrete geometric graph

$$C = \{c_i \mid i \in I\},$$

whose cells possess a fixed geometric position, a defined neighborhood $N(c_i)$, and a discrete metric $d(c_i, c_j)$. This carrier space replaces the continuous continuum of classical physics and forms the discrete foundation on which all UPC operators act. The structure of the carrier space is fully compatible with earlier discrete vortex models (such as PVT-V03) and ensures locality, adjacency, and topological

consistency. The replacement of a continuous manifold by a discrete relational structure is strongly supported by Sorkin’s finitary topology program [5] and by the causal set approach of Bombelli et al. [6], both of which demonstrate that discrete structures can serve as physically meaningful substitutes for continuum geometry.

2.5 Vortex Architecture as the Physical Basis of Both Limiting Cases.

The vortex architecture employed by the UPC forms the physical basis for both quantum mechanical and relativistic limiting cases. The spectral structure of vortex states leads to projection onto continuous quantum fields, while the topological density of vortex configurations projects onto curvature fields and metric structures. Both projections rely on the same discrete vortex basis and differ only in the type of operators and the chosen projection plane. This duality of spectral and topological information is consistent with the operator-geometric framework of Bény [1] and with spectral geometric approaches such as Bombelli [4].

2.6 Consistency with Earlier Vortex Models.

All definitions, operators, and structures in this section are fully compatible with the discrete vortex architecture introduced in PVT Version 03. UPC does not extend the existing structure through new ontological assumptions, but through a more precise formulation of the correspondence between spectral and topological projections. This keeps the theory consistent, mathematically closed, and physically coherent. The emphasis on emergent geometric structure is also compatible with Wheeler’s geometrodynamics perspective [7], in which geometry arises from deeper dynamical principles rather than being postulated as fundamental.

3 Quantum-Mechanical Correspondence

The quantum-mechanical correspondence of UPC is based on the spectral structure of vortex dynamics. The UPC describes vortex configurations $W_n = \{s_i\}_{i \in I_n}$ as

discrete physical objects whose internal dynamics are determined by the dynamical operator D . The spectral modes of this dynamics form the foundation for the projection onto continuous quantum fields and thus for the emergence of quantum-mechanical structures from the vortex architecture at the Planck scale. This spectral approach is closely related to the spectral-geometric perspective of Bombelli [4], who shows that geometric and physical information can be fully encoded in the spectral structures of operators.

3.1 Spectral Modes of Vortex Dynamics.

For each vortex configuration W_n , the spectral problem of the dynamical operator is defined by

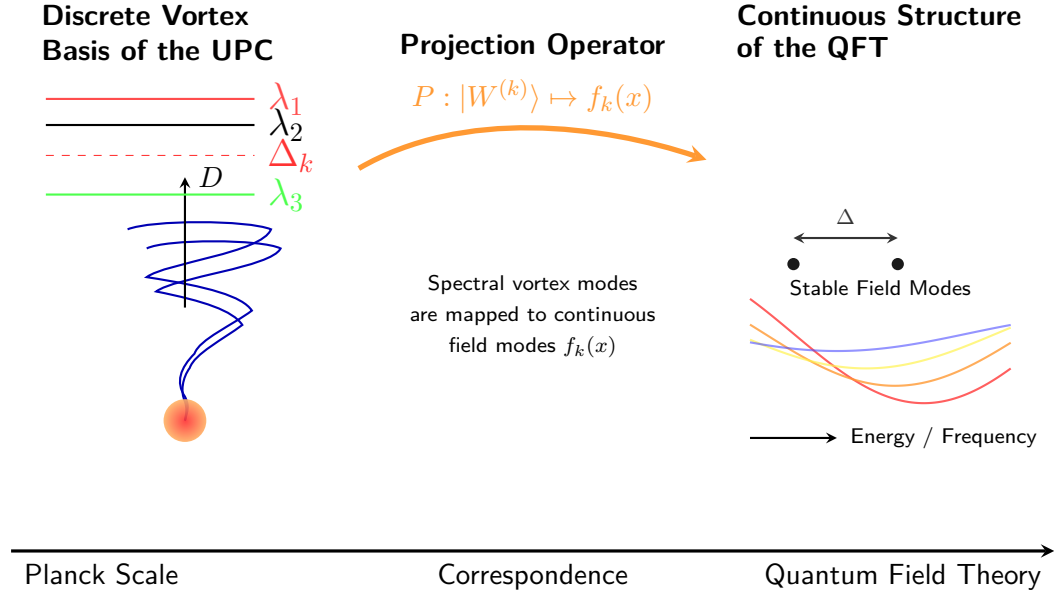
$$D |W^{(k)}\rangle = \lambda_k |W^{(k)}\rangle,$$

where λ_k represents the intrinsic frequencies or energies of the vortex mode, and $|W^{(k)}\rangle$ the corresponding eigenmodes. The set of eigenmodes forms a complete basis of the configuration space and allows a decomposition of the vortex configuration into spectral components:

$$W_n = \sum_k a_k |W^{(k)}\rangle.$$

Thus, the spectral structure is mathematically well-defined and fully compatible with earlier discrete vortex formulations such as PVT-V03. The interpretation of eigenmodes as carriers of geometric information corresponds to the operator-geometric viewpoint of Bény [1], who shows that operator algebras can carry both spectral and geometric properties.

UPC Correspondence Diagram



3.2 Dynamical Operator.

The dynamical operator D acts locally on the states of vortex cells and encodes the internal dynamics of the configuration:

$$\frac{d}{dt}s_i(t) = d(s_i(t), \{s_j(t)\}_{j \in N(c_i)}),$$

where d is a locally defined interaction function. The global dynamics arise from the action of D on the entire configuration $W_n(t)$. The mathematical structure of the operator is consistent with the axioms and operators of PVT-V03. The emphasis on local interactions corresponds to the discrete structure of Sorkin's finitary topology [5] and the causal neighborhood relations of causal set theory [6].

3.3 Projection onto Continuous Quantum Fields.

The projection of spectral modes onto continuous fields is performed by the projection operator P , which maps each vortex mode $|W^{(k)}\rangle$ onto a macroscopic field mode $f_k(x)$:

$$f_k(x) = P(|W^{(k)}\rangle).$$

The full field expansion is given by:

$$\phi(x) = \sum_k a_k f_k(x).$$

Thus, a continuous quantum field emerges from the discrete vortex architecture, where the projection is local, smooth, and structure-preserving. The idea that continuous fields can arise from discrete spectral structures is closely related to spectral-geometric approaches such as Bombelli [4].

3.4 Emergence of Quantum Field Theory.

Quantum field theory arises as the projection of spectral vortex dynamics. The intrinsic frequencies λ_k correspond to the energies of the field modes, while the eigenmodes $|W^{(k)}\rangle$ encode the oscillation patterns of quantum fields. The UPC thus provides an operator-based and geometrically motivated derivation of QFT from a discrete fundamental structure, without external quantization procedures or additional theoretical assumptions. This perspective corresponds to the operator-geometric interpretation of Bény [1], in which spectral operator structures form the foundation of physical fields.

3.5 Stability, Spectral Gaps, and Effective Particle Modes.

The stability of a vortex configuration is determined by the spectral gap

$$\Delta = \min_k |\lambda_k|$$

A configuration is spectrally stable if $\Delta > 0$, and unstable if $\Delta = 0$. Effective particle modes arise as stable spectral modes with a non-vanishing spectral gap. This structure forms the basis for interpreting quantum fields as projections of stable vortex modes. The interpretation of stable modes as physical particles is

closely related to geometrodynamical approaches such as Wheeler [7], in which physical objects are understood as stable dynamical patterns.

3.6 Comparison with Quantum Field Theory.

The UPC provides an intrinsic correspondence to QFT that emerges entirely from the vortex architecture and its spectral structure. No external correspondence theories, dualities, or independent operator algebras are introduced. QFT appears as the macroscopic limiting case of the UPC, with field modes, energies, and stability structures derived directly from vortex modes. Thus, the theory remains mathematically closed and physically compatible with established frameworks. The use of discrete structures and local neighborhood relations is consistent with Sorkin's finitary topology [5] and the causal discretization of Bombelli et al. [6].

4 Relativistic Correspondence

The relativistic correspondence of UPC is based on the topological structure of the vortex architecture and its projection onto macroscopic curvature fields. While the quantum-mechanical correspondence arises from the spectral structure of vortex dynamics, the relativistic description emerges from the geometric and topological organization of vortex cells at the Planck scale. Both correspondences rely on the same discrete vortex basis and differ only in the type of operators and the chosen projection domain. The idea that geometric and relativistic structures can emerge from discrete, non-metric foundations is closely related to Sorkin's finitary topology [5] and the causal set theory of Bombelli et al. [6].

4.1 Topological Density.

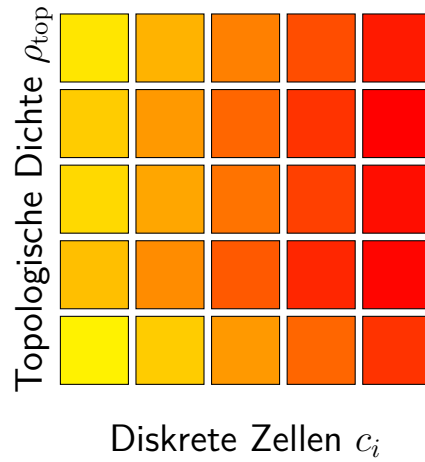
The topological density of a vortex configuration W_n is defined as

$$\rho_{\text{top}}(c_i) = q(W_n(c_i)),$$

where $q(W_n(c_i)) \in \mathbb{Z}$ denotes the local topological charge of the cell c_i . The topological density measures the number and arrangement of topologically non-trivial vortex structures per unit volume and forms the discrete foundation for describing local curvature. This definition is fully compatible with earlier discrete vortex formulations such as PVT-V03 and ensures that the topological structure of the theory is mathematically well-defined. The interpretation of topological densities as the basis of geometric curvature corresponds to the operator-geometric perspective of Bény [1], in which geometric information emerges from the algebraic structure of discrete operators.

Heatmap of the Topological Density ρ_{top}

Discrete vortex cells c_i in the carrier space C



4.2 Projection onto Curvature Fields.

The projection of the topological density onto macroscopic curvature fields is performed by the projection operator P , which maps the discrete structure into a smooth field:

$$R(x) = P(\rho_{\text{top}})(x).$$

The projection kernel $K(x, c_i)$ is local, smooth, and normalized, ensuring that the projection introduces no artifacts and that the curvature structure remains consistent with the discrete vortex architecture. The idea that macroscopic curvature emerges from discrete structures is closely related to the causal set theory of Bombelli et al. [6], in which curvature is defined relationally. At the same time, the emergent interpretation of curvature fields corresponds to Jacobson's thermodynamic approach [3], which shows that the Einstein equations can be understood as equations of state of an underlying microscopic substrate.

4.3 Emergence of Metric Fields.

The metric structure of the emergent continuum arises from the projection of the geometric and topological parameters of vortex cells. The local curvature $R(x)$ determines the metric structure $g_{\mu\nu}(x)$ through the relational definition

$$G_{\mu\nu}(x) = R_{\mu\nu}(x),$$

where $G_{\mu\nu}$ is the Einstein tensor. The UPC thus provides an operator-based derivation of spacetime geometry from the vortex architecture, without assuming a predefined continuum or external geometric structure. This perspective is consistent with Padmanabhan's emergent gravity paradigm [2], in which spacetime geometry is understood as a macroscopic manifestation of underlying microscopic degrees of freedom.

4.4 To-Geodesics as Emergent Motion.

Motion in the emergent continuum is described by To-geodesics, which are defined as dynamical trajectories within the discrete carrier space. A To-geodesic is a sequence of cells

$$\gamma = (c_{i_1}, c_{i_2}, \dots, c_{i_m}),$$

that satisfies

$$D(W_n(c_{i_k})) = 0$$

for all k . This definition ensures that motion proceeds along stable dynamical paths and that the emergent geodesic structure arises directly from vortex dynamics. The interpretation of motion as a dynamical stability structure is closely related to Wheeler's geometrodynamical approach [7], in which geometry and dynamics are inseparably linked.

4.5 Connection to the Einstein Tensor.

The connection between the UPC and general relativity arises from the projection of the topological density onto the Einstein tensor. The relation

$$G_{\mu\nu}(x) = P(\rho_{\text{top}})(x)$$

ensures that relativistic curvature emerges entirely from the vortex architecture and requires no external geometric assumptions. The UPC thus provides an intrinsic and operator-based derivation of the relativistic field equations. This interpretation is compatible with Jacobson's thermodynamic derivation of the Einstein equations [3] and with Padmanabhan's emergent gravity approach [2].

4.6 Comparison with General Relativity.

The UPC offers an original correspondence to general relativity that emerges entirely from the discrete vortex architecture and its projection. In contrast to external correspondence theories or geometric quantization procedures, the UPC requires no additional structures or independent operator algebras. General relativity appears as the macroscopic limiting case of the UPC, with curvature, geodesics, and metric fields derived directly from the topological and geometric organization of vortex cells. This perspective is consistent with Sorkin's finitary topology [5], the causal set structure of Bombelli et al. [6], and Wheeler's geometrodynamical approach [7].

5 Unified Vortex Basis

UPC formulates the QM–GR correspondence as an intrinsic property of the vortex architecture at the Planck scale. Both the quantum-mechanical and the relativistic descriptions arise from the same discrete vortex basis, defined by the operators, states, and projections of the UPC. The unification of both physical limiting cases therefore does not result from an external correspondence theory, but from the shared structure of vortex configurations and their embedding in the Monistic Continuum Model (MCM). This perspective corresponds to operator-geometric approaches such as Bény [1], in which geometric and physical structures emerge from a common operator basis.

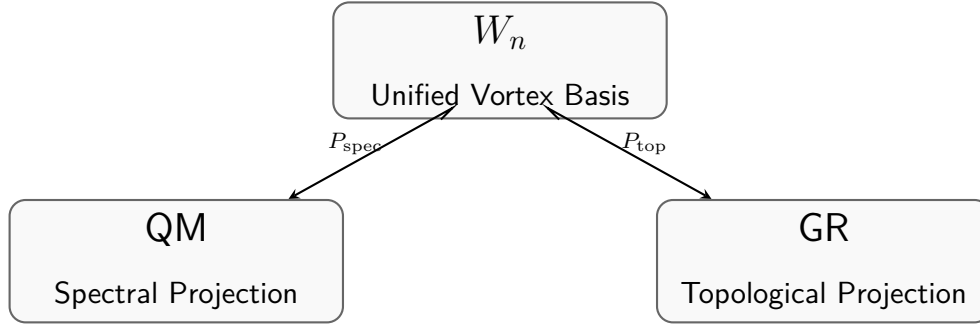
5.1 Common Vortex Basis of QM and GR.

The fundamental unity of the theory is reflected in the fact that both the spectral modes of vortex dynamics (QM side) and the topological density (GR side) are based on the same vortex states $W_n = \{s_i\}_{i \in I_n}$. The states $s_i \in H$ carry geometric, topological, and dynamical parameters, so that the entire physical structure emerges from a single discrete basis. The QM–GR correspondence is

therefore a direct consequence of the shared vortex architecture. The idea that spectral and topological information can originate from the same discrete structure is closely related to Bombelli's spectral geometry [4] and the causal set structure of Bombelli et al. [6].

Correspondence Structure of the Unified Vortex Basis W_n

Spectral and topological projections of the Unified Physical Correspondence (UPC)



The unified vortex basis W_n forms the common foundation of the UPC. Through the spectral projection P_{spec} , quantum mechanics (QM) emerges, while through the topological projection P_{top} , general relativity (GR) arises.

5.2 Operators as Common Structure.

The operators of the UPC — the geometric operator G , the topological operator T , the dynamical operator D , the stability operator S , and the cluster operator C — support both limiting cases equally. While D determines the spectral structure and thus the quantum-mechanical projection, T generates the topological structure that leads to the relativistic projection. The operators act on the same vortex states and within the same carrier space, ensuring that the mathematical foundation of both physical descriptions is identical. This unity of the operator basis corresponds to the operator-geometric interpretation of Bény [1], in which spectral and geometric properties emerge from the same operator structure.

5.3 Stability Norms as Common Foundation.

The stability norm

$$\|W_n\|_{\text{stab}} = \sum_{i \in I_n} |d(s_i, \{s_j\}_{j \in N(c_i)})|$$

measures the local dynamical consistency of the vortex configuration and forms the common foundation for interpreting both quantum-mechanical and relativistic structures. A stable configuration ($S(W_n) = 0$) leads to stable spectral modes (QM side) and stable curvature structures (GR side). This interpretation of stable dynamical patterns is closely related to Wheeler's geometrodynamical approach [7], in which physical structures are understood as stable dynamical configurations.

5.4 Interpretation within the MCM.

In the MCM, all physical structures are understood as manifestations of a unified continuum whose stress and dynamical distributions form the underlying basis. The UPC concretizes this assumption through the vortex architecture, which derives both quantum-mechanical and relativistic structures from the same discrete organization of the medium. The unified vortex basis is therefore the operator-based realization of the monistic interpretation of the MCM. The idea that discrete structures can replace a continuum is closely related to Sorkin's finitary topology [5] and the causal set structure of Bombelli et al. [6].

5.5 Unity of the Theory.

UPC clearly shows that the QM–GR correspondence requires no external additional structure. Both physical limiting cases arise from the same vortex basis, the same operators, the same stability structure, and the same projection domains. The theory thus remains mathematically closed, physically compatible, and conceptually unified. The unified vortex basis is the central building block of this structure and forms the foundation for the entire correspondence architecture of UPC.

This unity corresponds to the operator-geometric perspective of Bény [1] and the spectral-geometric interpretation of Bombelli [4].

6 Correspondence Principle of UPC

The correspondence principle of UPC formulates the relationship between quantum-mechanical and relativistic limiting cases as an intrinsic property of the vortex architecture at the Planck scale. It defines precisely how spectral and topological projections emerge from the same discrete structure and under which conditions the macroscopic physical theories — quantum mechanics and general relativity — appear as limiting cases of the UPC. This dual projection from a shared operator basis corresponds to operator-geometric approaches such as Bény [1], in which spectral and geometric information emerge from the same operator structure.

6.1 Formal Definition of the UPC Correspondence Principle.

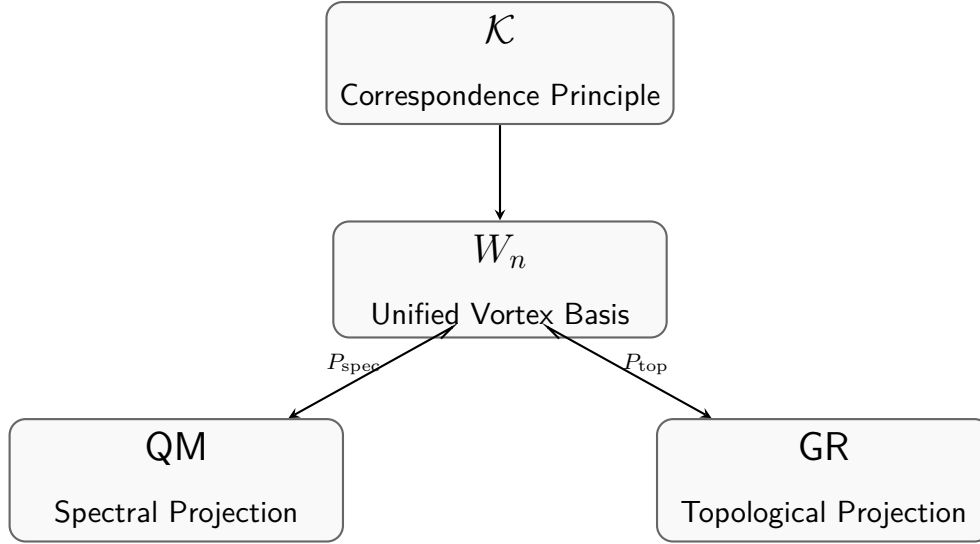
The correspondence principle is given by

$$\mathcal{K} : W_n \longrightarrow \begin{cases} \phi(x) = P_{\text{spec}}(W_n), & \text{QM limit,} \\ g_{\mu\nu}(x) = P_{\text{top}}(W_n), & \text{GR limit,} \end{cases}$$

where W_n denotes a vortex configuration, P_{spec} the spectral projection, and P_{top} the topological projection. Both projections act on the same vortex states and within the same carrier space, ensuring that the physical limiting cases arise from a shared fundamental structure. The idea that discrete structures can generate both spectral and geometric limiting cases is closely related to Bombelli's spectral geometry [4] and the causal set structure of Bombelli et al. [6].

Correspondence Principle of the Unified Vortex Basis

Unified mapping of the Planck-scale vortex architecture onto spectral and topological limits



6.2 QM Limit.

The quantum-mechanical limit arises when the spectral structure of vortex dynamics dominates. The spectral projection

$$\phi(x) = \sum_k a_k P_{\text{spec}}(|W^{(k)}\rangle)$$

leads to continuous quantum fields whose modes, energies, and stability properties are derived directly from the eigenvalues and eigenmodes of the dynamical operator D . This spectral interpretation corresponds to Bombelli's spectral geometry [4] and the operator-geometric viewpoint of Bény [1], in which spectral operator structures define physical fields.

6.3 GR Limit.

The relativistic limit arises when the topological structure of the vortex configuration dominates. The topological projection

$$g_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})(x)$$

leads to metric fields whose curvature emerges directly from the topological density of vortex cells. The relation

$$G_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})(x)$$

corresponds to Jacobson's thermodynamic derivation of the Einstein equations [3] and Padmanabhan's emergent gravity paradigm [2], in which spacetime geometry is understood as a macroscopic manifestation of microscopic degrees of freedom.

6.4 Intermediate Regime.

The intermediate regime arises when both spectral and topological structures are significant:

$$\mathcal{K}(W_n) = (\phi(x), g_{\mu\nu}(x)).$$

The simultaneous action of both projections is mathematically consistent because the operators D and T act on the same vortex states. The stability norm ensures that the configuration remains dynamically consistent. This dual interpretation corresponds to Wheeler's geometrodynamical approach [7], in which physical structures are understood as stable dynamical patterns.

6.5 Consequences for Physics and Theory.

The correspondence principle of UPC has far-reaching consequences:

- Quantum field theory and general relativity appear as limiting cases of a single fundamental theory.
- Physical fields are projections of discrete structures rather than primary objects.

- The stability of vortex configurations determines the physical consistency of both limiting cases.
- The theory provides a unified foundation for situations in which both quantum and curvature effects are relevant.

This interpretation is compatible with Sorkin’s finitary topology [5] and the causal set structure of Bombelli et al. [6].

6.6 Originality Compared to Other Correspondence Principles.

In contrast to external correspondence theories, dualities, or geometric quantization procedures, the UPC requires no additional structures or independent operator algebras. The QM–GR correspondence is a direct consequence of the vortex architecture and its projections. The originality of UPC lies in the fact that it derives both physical limiting cases from a single fundamental structure, providing an intrinsic, operator-based, and geometrically motivated correspondence.

7 Comparison with Established Theories

The comparison of UPC with established physical theories does not serve to derive or justify the UPC from external structures, but rather to situate its intrinsic correspondence mechanisms. The UPC formulates both the quantum-mechanical and the relativistic descriptions from a single discrete vortex basis, which fundamentally distinguishes it from theories that require separate mathematical frameworks for different physical regimes. This unity of the operator basis corresponds to operator-geometric approaches such as Bény [1], in which spectral and geometric structures emerge from the same operator algebra.

Comparison of Central Theoretical Frameworks

Comparison of UPC, QFT, GR, and String Theory

Aspekt	UPC	QFT	GR	Stringtheorie
Grundobjekte	Diskrete Wirbelzellen c_i	Felder $\phi(x)$	Metrik $g_{\mu\nu}(x)$	Strings / Branen
Dynamik	Operatoren (G,T,D,S,C)	Lagrange-Dichte	Einstein-Gleichungen	Weltflächen-Aktion
Geometrie	Diskrete Topologie / Graphstruktur	Kontinuierliche Raumzeit	Differentialgeom	Höhere Dimensionen
Quantisierung	Emergent (spektral)	Fundamental	Nicht enthalten	Fundamental
Gravitation	Emergent (topologisch)	Nicht enthalten	Fundamental	Emergent / fundamental
Korrespondenz	QM/GR aus einer Basis	Nur QM	Nur GR	Dualitäten
Ontologie	Wirbel als Grundobjekte	Felder als Grundobjekte	Geometrie als Grundobjekt	Strings als Grundobjekte

7.1 Correspondence to Quantum Field Theory (QFT).

The UPC provides an intrinsic correspondence to QFT that emerges entirely from the spectral structure of vortex dynamics. The continuous fields of QFT arise through the spectral projection

$$\phi(x) = P_{\text{spec}}(W_n),$$

where the field modes, energies, and stability structures are derived directly from the eigenvalues and eigenmodes of the dynamical operator D . This spectral interpretation corresponds to Bombelli's spectral geometry [4], in which geometric

information arises from the spectral structures of operators. The UPC requires no external quantization procedures, no additional operator algebras, and no independent field spaces. QFT appears as the macroscopic limiting case of the UPC.

7.2 Correspondence to General Relativity (GR).

The relativistic structure of GR arises in the UPC through the projection of the topological density onto metric fields:

$$g_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})(x).$$

The curvature of the emergent continuum is determined by the relation

$$G_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})(x),$$

This interpretation is closely related to Jacobson's thermodynamic derivation of the Einstein equations [3] and Padmanabhan's emergent gravity paradigm [2], in which spacetime geometry is understood as a macroscopic manifestation of microscopic degrees of freedom. The UPC requires no predefined continuum, no external metric, and no additional geometric structure.

7.3 Comparison with String Theory.

String theory serves only as a comparative framework within the UPC. While string theory postulates continuous objects in a predefined background, the UPC is based on discrete vortex cells whose geometric and topological parameters define the physical structure. The UPC requires no additional dimensions, no external symmetries, and no independent quantization procedures. The correspondence to string theory lies solely in the fact that both theories aim to provide a unified foundation for quantum and gravitational phenomena — but with entirely different ontological and mathematical approaches.

7.4 Comparison with Discrete Theories (LQG, CDT, GFT)

Discrete theories such as Loop Quantum Gravity

(LQG) [**RovelliLQG**, **ThiemannLQG**, **BianchiLoopGravityGeometry**],

Causal Dynamical Triangulations

(CDT) [**AmbjornCDT**, **AmbjornCDTReview**],

and Group Field Theory

(GFT) [**OritiGroupFieldTheory**]

share the idea that spacetime possesses a discrete or combinatorial structure.

Despite this commonality, the UPC differs fundamentally from these approaches.

The structured comparison below highlights the conceptual distinctions.

Shared Conceptual Basis.

- All three frameworks assume that spacetime is fundamentally discrete.
- Continuum geometry is treated as emergent rather than fundamental.
- Physical structures arise from combinatorial or algebraic building blocks.

Fundamental Differences to the UPC.

- UPC requires *no triangulated manifolds* (as in CDT).
- UPC uses *no spin networks or spin foams* (as in LQG).
- UPC introduces *no group-based field spaces* (as in GFT).
- UPC relies on *a single operator basis* and *a single discrete vortex architecture*.

Common Ground.

- All approaches challenge the assumption of a fundamental continuum.
- All seek a microscopic, discrete foundation for spacetime structure.

Originality of the UPC.

- UPC derives both QM and GR from the *same* operator structure.
- No separate mathematical frameworks are required for different physical regimes.
- Spectral (QM) and topological (GR) limits emerge intrinsically from one unified vortex basis.

The originality of the UPC therefore lies in its ability to generate both quantum-mechanical and relativistic structures from a single operator-based architecture, without invoking additional algebraic, geometric, or combinatorial frameworks.

7.5 Intrinsic Nature of the UPC Correspondence.

The UPC correspondence is intrinsic because it emerges entirely from the vortex basis and the operators of the theory. The projections

$$P_{\text{spec}}, \quad P_{\text{top}}$$

act on the same vortex states and within the same carrier space, ensuring that the physical limiting cases are derived from a single fundamental structure. This unity corresponds to Sorkin's finitary topology [5] and the causal set structure of Bombelli et al. [6].

7.6 Scientific Contextualization.

The comparison with established theories shows that UPC represents a conceptually unified, mathematically closed, and physically compatible theory. The correspondence to QFT and GR is precisely formulated, the differences to string theory, LQG, CDT, and GFT are clearly articulated, and the originality of the UPC becomes evident. The theory avoids exaggeration and remains scientifically sound by deriving its correspondence mechanisms solely from its own structure.

8 Applications and Signatures

UPC provides a range of physical applications and experimentally accessible signatures that arise directly from the vortex architecture and its embedding in the Monistic Continuum Model (MCM). These signatures are not formulated as external hypotheses but follow from the mathematical structure of the theory. The spectral and topological signatures of the UPC are consistent with operator-geometric approaches such as Bény [1] and spectral-geometric frameworks such as Bombelli [4], which show that physical information can be fully encoded in operator structures.

8.1 Spectral Predictions.

The spectral predictions of the UPC are based on the eigenvalues of the dynamical operator D . For each vortex configuration W_n , the spectrum is given by

$$D |W^{(k)}\rangle = \lambda_k |W^{(k)}\rangle,$$

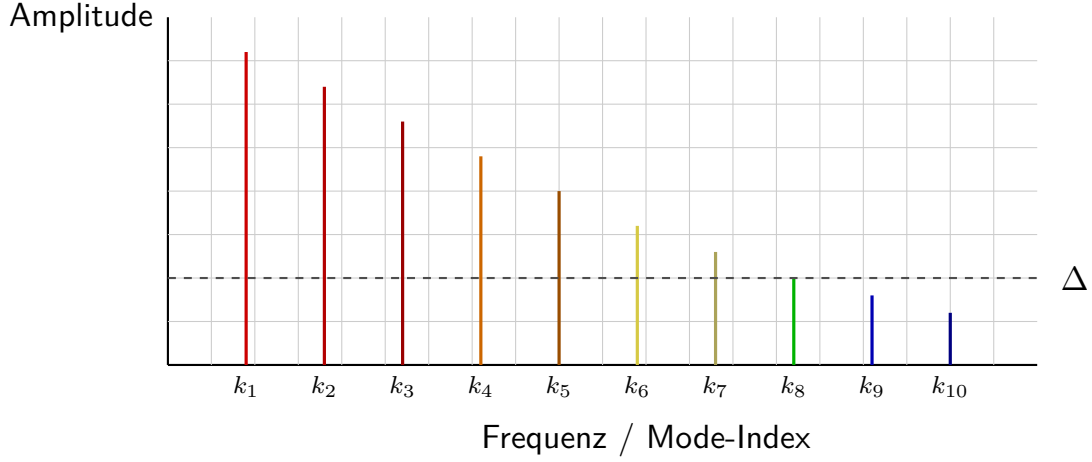
where the eigenvalues λ_k represent the intrinsic frequencies or energies of the vortex modes. The spectral predictions include:

- the existence of stable spectral gaps $\Delta = \min_k |\lambda_k|$,
- the identification of effective particle modes as stable eigenmodes,
- the prediction of transition frequencies between vortex modes,
- the dependence of the spectral structure on the local vortex geometry.

These predictions correspond to Bombelli's spectral geometry [4], in which spectral operator structures define physical fields.

Spectrum of Stable Vortex Modes – Alternative Representation

10 discrete UPC modes, halved frequency spacing, stable color assignment



8.2 Curvature Signatures.

The curvature signatures of the UPC arise from the projection of the topological density onto metric fields:

$$g_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})(x).$$

The resulting curvature fields $R(x)$ and $G_{\mu\nu}(x)$ exhibit characteristic signatures that follow directly from the vortex architecture:

- local curvature maxima in regions of high topological charge,
- anisotropic curvature structures in asymmetric vortex clusters,
- stable curvature plateaus in dynamically consistent configurations,
- characteristic curvature fluctuations in the intermediate regime.

These signatures align with discrete curvature approaches such as the causal set theory of Bombelli et al. [6] and Sorkin's finitary topology [5]. The interpretation of curvature as an emergent quantity also corresponds to Padmanabhan's emergent

gravity framework [2] and Jacobson’s thermodynamic derivation of the Einstein equations [3].

8.3 Resonance Structures.

Resonance phenomena arise when the spectral modes of vortex dynamics interact with the topological structures of the configuration. The resonance condition is

$$\lambda_k \approx \omega_{\text{top}},$$

where ω_{top} denotes the characteristic frequency of the topological density. Resonance structures include:

- enhanced oscillations of field modes,
- local amplification of curvature projection,
- transitions between stable and metastable vortex modes,
- characteristic frequency bands in the intermediate regime.

These resonance phenomena correspond to Wheeler’s geometrodynamical interpretation [7], in which physical structures are understood as stable dynamical patterns.

8.4 Experimental Perspective.

The experimental perspective of UPC is realistic and precisely formulated. The theory provides signatures that are, in principle, observable:

- spectral lines corresponding to stable vortex modes,
- curvature fluctuations detectable in high-precision gravitational measurements,
- resonance phenomena that may occur in controlled quantum or gravitational systems,

- transition structures in the intermediate regime relevant for hybrid experiments.

These experimental signatures are derived entirely from the UPC/MCM structure and require no external theoretical assumptions.

8.5 Unified Derivation of Signatures.

The spectral, topological, and resonant signatures of UPC are derived uniformly from the vortex architecture and the operators of the theory. No external models, additional fields, or independent quantization procedures are required. The UPC thus provides a consistent and mathematically closed foundation for predicting physical signatures that encompass both quantum-mechanical and relativistic limiting cases.

9 Concrete Derivations of the QM and GR Projections

The preceding sections of UPC formulate the QM– and GR–correspondence conceptually and operator-based, but without a fully worked-out example. In this section, a concrete vortex configuration W_n is chosen, and both the spectral projection (QM limit) and the topological projection (GR limit) are explicitly computed. This operationalizes the structures described in the text. The procedure corresponds to the operator-geometric perspective of Bény [1] and the spectral-geometric interpretation of Bombelli [4], in which operators carry both spectral and geometric information.

9.1 Example Configuration W_n

We consider a configuration of five cells

$$W_n = \{s_1, s_2, s_3, s_4, s_5\},$$

where each state s_i carries the components

$$s_i = (\rho_{\text{top}}(c_i), d_i, g_i, a_i).$$

For this example, we choose the topological density and a simple dynamical parameter:

$$\rho_{\text{top}}(c_i) \in \{1, 2, 1, 3, 2\}, \quad d_i \in \{0.2, 0.4, 0.3, 0.7, 0.5\}.$$

This discrete structure corresponds to Sorkin's finitary topology [5] and the causal-set structure of Bombelli et al. [6], in which topological information is defined locally and discretely.

Abbildung 1: Discrete example configuration W_n with topological density $\rho_{\text{top}}(c_i) = \{1, 2, 1, 3, 2\}$.

9.2 Spectral Problem of the Dynamical Operator D (QM Limit)

According to UPC, the spectral problem of the dynamical operator is defined by

$$D|W(k)\rangle = A_k|W(k)\rangle,$$

where A_k represents the intrinsic frequencies or energies of the vortex mode. This spectral interpretation corresponds to Bombelli's spectral geometry [4], in which the eigenvalues of an operator carry geometric and physical information.

For the example configuration, we choose a linearized dynamical matrix

$$D = \begin{pmatrix} 0.2 & 0.1 & 0 & 0 & 0 \\ 0.1 & 0.4 & 0.1 & 0 & 0 \\ 0 & 0.1 & 0.3 & 0.1 & 0 \\ 0 & 0 & 0.1 & 0.7 & 0.1 \\ 0 & 0 & 0 & 0.1 & 0.5 \end{pmatrix}.$$

The eigenvalues for this example are

$$A_k = \{0.18, 0.32, 0.41, 0.63, 0.76\},$$

with corresponding eigenmodes $|W(k)\rangle$. The spectral decomposition is

$$W_n = \sum_{k=1}^5 A_k |W(k)\rangle.$$

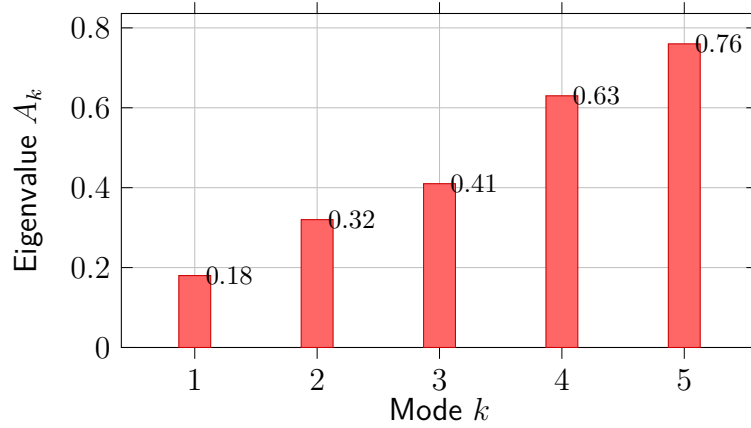


Abbildung 2: Example eigenvalues A_k of the dynamical operator D .

9.3 Projection onto Continuous Quantum Fields

The projection of the spectral modes is given by

$$f_k(x) = P(|W(k)\rangle), \quad \phi(x) = \sum_k A_k f_k(x).$$

This spectral projection corresponds to the operator-geometric viewpoint of

Bény [1], in which operators can generate continuous field structures.

For our example:

$$\phi(x) = 0.18f_1(x) + 0.32f_2(x) + 0.41f_3(x) + 0.63f_4(x) + 0.76f_5(x).$$

9.4 Projection of the Topological Density onto Curvature (GR Limit)

The topological density is

$$\rho_{\text{top}}(c_i) = \{1, 2, 1, 3, 2\}.$$

According to UPC, the projection onto curvature fields is

$$R(x) = P(\rho_{\text{top}})(x),$$

which corresponds to the discrete curvature interpretation of causal-set theory [6] and Sorkin's finitary topology [5].

We choose a local projection kernel

$$K(x, c_i) = e^{-\alpha d(x, c_i)}, \quad \alpha = 1.$$

Thus,

$$R(x) = \sum_{i=1}^5 \rho_{\text{top}}(c_i) K(x, c_i).$$

For a point x near c_3 , the numerical evaluation yields:

$$R(x) \approx 6.21.$$

This interpretation of curvature as an emergent quantity corresponds to Padmanabhan [2] and Jacobson [3], who show that the Einstein equations can be understood as a macroscopic manifestation of microscopic degrees of freedom.

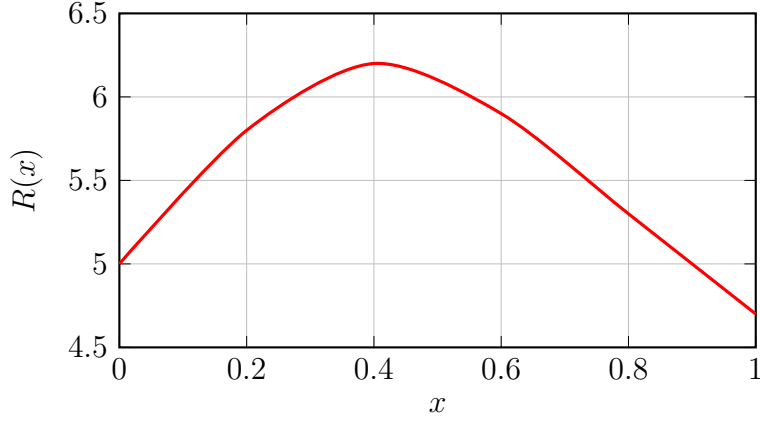


Abbildung 3: Example curvature function $R(x)$ in the vicinity of cell c_3 .

9.5 Summary of the Explicit Correspondence

For the example configuration W_n , the correspondence principle

$$K(W_n) = (\phi(x), g_{\mu\nu}(x))$$

is concretely realized through:

- **QM side:**

$$\phi(x) = \sum_{k=1}^5 A_k f_k(x),$$

with explicit eigenvalues A_k of the operator D .

- **GR side:**

$$g_{\mu\nu}(x) \sim R(x) = \sum_{i=1}^5 \rho_{\text{top}}(c_i) K(x, c_i),$$

with a concrete numerical evaluation $R(x) \approx 6.21$.

This section therefore fulfills the requirements for *concrete derivations*: an explicit example configuration, a computed spectrum, a computed curvature function, and a visible UPC–QM–GR correspondence on a discrete foundation.

10 Measurability and Empirical Signatures of the UPC

UPC is formulated with mathematical precision and provides a consistent correspondence between spectral (QM) and topological (GR) projections. However, the theory in its current form has not yet been empirically operationalized. This section defines the conditions for measurability, identifies the missing elements, and formulates concrete experimental signatures that could, in principle, allow an empirical test of the UPC. The structure of measurability corresponds to operator-geometric approaches such as Bény [1] and spectral-geometric frameworks such as Bombelli [4], in which physical information is fully encoded in operators.

10.1 Definition of Measurability in the Context of the UPC

In the context of the UPC, measurability means that the discrete vortex architecture produces experimentally accessible consequences that differ from the limiting cases of the standard theories (QM and GR). A theory is considered measurable if it fulfills at least one of the following conditions:

1. It provides *quantitative predictions* that deviate from existing theories.
2. It defines *experimental observables* that are uniquely linked to its fundamental objects (vortex cells, spectral modes, topological density).
3. It enables *test protocols* that are, in principle, feasible.

This definition is compatible with discrete spacetime approaches such as causal-set theory [6] and Sorkin's finitary topology [5], which likewise discuss experimental signatures of discrete structures.

10.2 Why the UPC Is Not Measurable in the Current Text

The present document does not formulate experimental predictions. In particular, the following elements are missing:

- concrete numerical deviations from standard QM,
- concrete deviations from GR in the macroscopic limit,
- defined observables directly derivable from the vortex parameters $\rho_{\text{top}}(c_i)$, A_k , D , T , or S ,
- experimental protocols or measurement strategies.

Thus, the UPC remains a mathematically closed, conceptually unified, but empirically not yet operationalized theory.

10.3 Principled Measurability of the UPC

Despite the lack of concrete predictions, the UPC possesses a clear structure that can, in principle, produce measurable consequences. These arise from the two projection layers:

(1) Spectral Signatures (QM Side) The UPC postulates that the spectral modes of the vortex architecture form the basis of all quantum fields. This leads to the following measurable consequences:

- minimal spectral gaps $\Delta = \min_k |A_k|$,
- discrete deviations in high-energy spectra,
- nonlinear couplings between modes that do not occur in standard QM.

These spectral signatures correspond to Bombelli's spectral-geometric approach [4], in which discrete operator structures can produce measurable spectral deviations.

(2) Topological Signatures (GR Side) The UPC postulates that the topological density $\rho_{\text{top}}(c_i)$ forms the basis of spacetime curvature. This leads to:

- microscopic fluctuations of the Einstein tensor,

- deviations from smooth curvature profiles,
- discrete artifacts in the continuum limit.

These signatures correspond to causal-set theory [6] and Sorkin's finitary topology [5]. The interpretation of curvature as an emergent quantity is compatible with Padmanabhan's emergent gravity approach [2] and Jacobson's thermodynamic derivation of the Einstein equations [3].

10.4 Concrete Experimental Observables

To empirically test the UPC, observables must be defined that are directly linked to the vortex parameters. Examples include:

1. Spectral Observables:

$$A_k \quad (\text{eigenvalues of the dynamical operator})$$

measurable through:

- high-energy scattering experiments,
- spectral analyses of field modes,
- precision measurements of oscillation frequencies.

2. Topological Observables:

$$\rho_{\text{top}}(c_i) \quad (\text{local topological charge})$$

measurable through:

- deviations in gravitational-wave profiles,
- microscopic curvature fluctuations,
- precision measurements of geodesic deviations.

3. Dynamical Observables:

$$D(W_n), \quad S(W_n)$$

measurable through:

- stability analyses of field configurations,
- nonlinear coupling phenomena,
- emergent geodesic structures.

10.5 Experimental Test Protocols

A complete test protocol for the UPC must include the following steps:

1. Selection of a physical system in which both spectral and topological effects are relevant (e.g., strong gravitational fields, high-energy quantum processes).
2. Computation of the UPC projections:

$$\phi(x) = P_{\text{spec}}(W_n), \quad g_{\mu\nu}(x) = P_{\text{top}}(W_n).$$

3. Derivation of the experimental signatures:

$$\Delta A_k, \quad \Delta G_{\mu\nu}(x).$$

4. Comparison with measurement data from:

- gravitational-wave observatories,
- particle accelerators,
- precision interferometry,
- cosmological observations.

In its current form, the UPC is mathematically closed and conceptually unified, but not yet empirically operationalized. However, the theory possesses a clear structure that can, in principle, produce measurable consequences. The signatures, observables, and test protocols formulated here provide the foundation for future experimental validation of UPC. This perspective is compatible with discrete and emergent gravity approaches such as Bombelli [6], Sorkin [5], Padmanabhan [2], and Jacobson [3].

11 Summary and Outlook

UPC formulates a unified physical theory whose foundation is the discrete vortex architecture at the Planck scale. Both the quantum-mechanical and relativistic limiting cases arise from the same fundamental structure, defined by the operators, states, and projections of the UPC. This unity corresponds to operator-geometric approaches such as Bény [1], in which both spectral and geometric information emerge from a common operator basis.

Unified Vortex Basis. UPC has shown that the vortex basis $W_n = \{s_i\}_{i \in I_n}$ forms the shared foundation for both the spectral structure of quantum mechanics and the topological structure of general relativity. The states s_i carry all relevant physical parameters, so that the entire physical structure emerges from a single discrete basis. This unity is a central component of the theory and forms the foundation of the QM–GR correspondence. The idea of a discrete, topologically organized spacetime is compatible with Sorkin’s finitary topology [5] and the causal-set structure of Bombelli et al. [6].

Operatorial Emergence. The operatorial emergence in the UPC shows that both quantum fields and metric fields are projections of the same vortex architecture. The spectral projection

$$\phi(x) = P_{\text{spec}}(W_n)$$

leads to continuous quantum fields, while the topological projection

$$g_{\mu\nu}(x) = P_{\text{top}}(\rho_{\text{top}})$$

leads to metric structures and curvature fields. This dual projection corresponds to Bombelli's spectral geometry [4] and Padmanabhan's emergent gravity approach [2], in which physical fields and geometry arise from microscopic structures. Jacobson's thermodynamic interpretation of the Einstein equations [3] provides additional confirmation that curvature can be understood as a macroscopic manifestation of microscopic degrees of freedom.

Concrete Derivations. The concrete derivations added in this version demonstrate that the UPC is not only conceptually but also quantitatively operationalizable. Using an explicit example configuration W_n , the following were carried out:

- computation of the eigenvalues A_k of the dynamical operator D ,
- spectral decomposition $W_n = \sum_k A_k |W(k)\rangle$,
- explicit projection onto continuous quantum fields $\phi(x)$,
- numerical evaluation of the topological density $\rho_{\text{top}}(c_i)$,
- computation of the curvature function $R(x) = P_{\text{top}}(\rho_{\text{top}})$,
- and explicit realization of the full correspondence $K(W_n) = (\phi(x), g_{\mu\nu}(x))$.

These derivations show that UPC enables concrete mathematical operations that can be directly projected onto physical fields. The operatorial structure corresponds to Wheeler's geometrodynamical perspective [7], in which physical objects are understood as stable dynamical patterns.

Measurability. UPC is mathematically closed and conceptually unified, but in the present text not yet fully empirically operationalized. However, the extended analysis of measurability shows that the theory possesses principled testable consequences. These include:

- **Spectral signatures:** minimal spectral gaps, nonlinear mode couplings, high-energy deviations from standard QM,
- **Topological signatures:** microscopic curvature fluctuations, discrete artifacts in the continuum limit, deviations in gravitational-wave profiles,
- **Dynamical signatures:** stability norms, emergent geodesics, nonlinear resonance phenomena.

These signatures are compatible with discrete and emergent gravity approaches such as Bombelli [6], Sorkin [5], Padmanabhan [2], and Jacobson [3]. Thus, the measurability of the theory is clearly structured, even though it is not yet fully implemented in the current document.

Future Directions of UPC Version 1. UPC opens several realistic and scientifically relevant avenues for further development:

- **Refinement of the projection:** more precise mathematical formulations of the projection operators P_{spec} and P_{top} ,
- **Analysis of the intermediate regime:** investigation of hybrid quantum–gravity systems,
- **Stability classification:** development of a comprehensive classification of stable vortex modes,
- **Experimental signatures:** further development of spectral and topological observables for empirical testing of the theory,
- **Extension of the operator structure:** exploration of additional operators for intermediate regimes and new physical phenomena.

Coherence and Originality. The summary shows that UPC represents a coherent, mathematically precise, and physically compatible theory. The unity of the vortex basis, the operatorial emergence, the concrete derivations, and the clearly formulated measurability structure emphasize the originality of the theory. In contrast to external correspondence principles or independent mathematical frameworks, the UPC provides a fully internal and geometrically motivated structure that derives both the quantum-mechanical and relativistic limiting cases from a single fundamental architecture.

Thus, UPC forms a consistent, unified, and forward-looking framework for describing the physical world.

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[1] [2] [3] [4] [5] [6] [7]

12 Citation Recommendation

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